PyCausalSim: A Python framework for causal discovery and effect estimation through simulation

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Software

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Summary

`PyCausalSim` is a Python framework for causal discovery and causal effect estimation built around a simulation-first workflow. Rather than treating causal discovery, effect estimation, and robustness checking as separate exercises in separate tools, `PyCausalSim` unifies them around structural causal models (SCMs) (Pearl, 2009) that can be discovered from data, specified by hand, simulated under interventions, and stress-tested against misspecification.

The core object, `CausalSimulator`, accepts observational data, discovers a candidate causal graph using one of several discovery algorithms — spanning constraint-based (PC, FCI), score-based (GES, FGES), functional (LiNGAM (Shimizu et al., 2006)), and continuous-optimization (NOTEARS (Zheng et al., 2018)) families — and exposes `simulate_intervention()`, a do-operator (Pearl, 2009) over the fitted structural model that returns interventional effect estimates with uncertainty intervals. Companion methods rank all causal drivers of a target (`rank_drivers()`) and search for intervention policies under constraints (`optimize_policy()`). A `StructuralCausalModel` class supports counterfactual generation with evidence conditioning. A validation module implements sensitivity analysis with confounding bounds, placebo tests, and refutation tests in the spirit of Sharma & Kiciman (2020). Applied modules provide experiment analysis with doubly-robust effect estimation (Bang & Robins, 2005) and heterogeneity analysis, uplift modeling (Gutierrez & Gérardy, 2017) with effect-based user segmentation, and marketing attribution using causal Shapley values with constrained budget optimization.

`PyCausalSim` is implemented on the scientific Python stack (`numpy`, `pandas`, `scipy`, `scikit-learn` (Pedregosa et al., 2011)), with optional adapter integrations to the wider causal ecosystem including DoWhy (Sharma & Kiciman, 2020) and EconML (Battocchi et al., 2019). It is released under the MIT license.

Statement of need

Applied analysts routinely need to answer interventional questions — "what happens to the outcome if we change this variable?" — from observational data. The dominant workflow answers correlational questions instead, because the causal toolchain is fragmented: graph discovery lives in one library, identification and estimation in another, sensitivity analysis in a third, and simulation of candidate structural models is typically hand-rolled. This fragmentation has practical consequences: analysts skip the robustness steps that don't fit the pipeline, and competing causal structures that are observationally indistinguishable — a mediator versus a confounder, for example — go untested even when they imply opposite decisions.

`PyCausalSim` addresses this gap with a single coherent API in which the simulation of structural models is the connective tissue: discovered or hypothesized graphs become generative objects that can be simulated under known interventions, compared against experimental or quasi-experimental evidence, and probed for the sensitivity of their conclusions. This simulation-first design also makes the framework suitable as a synthetic data-generating engine for downstream research — for example, generating training distributions of known causal environments for simulation-based inference (Cranmer et al., 2020) — a use case the author is actively developing for marketing measurement, where structural parameters such as advertising carryover and saturation must be inferred from short, confounded panels.

The intended audience is applied data scientists and computational researchers — particularly in marketing science, business analytics, and economics — who need decision-grade causal answers without assembling a bespoke toolchain, as well as researchers who need a programmable SCM simulation engine. Existing libraries each cover part of this surface: DoWhy (Sharma & Kiciman, 2020) provides an identification-and-refutation workflow, EconML (Battocchi et al., 2019) and CausalML (Chen et al., 2020) provide estimators for heterogeneous effects, and causal-learn provides discovery algorithms. `PyCausalSim` is complementary: it interoperates with these libraries through adapters while contributing the simulation layer — interventional simulation over discovered or specified SCMs, structure comparison under intervention, driver ranking, and policy search — as a first-class, unified workflow.

Functionality

The framework is organized around five capabilities:

- **Discovery.** Six causal discovery algorithms behind a common interface — PC and FCI (constraint-based), GES and FGES (score-based), LiNGAM (functional, for non-Gaussian data), and NOTEARS (continuous optimization) — returning candidate graphs that can be constrained with domain knowledge and visualized.
- **Simulation and intervention.** Fitted or user-specified SCMs are generative: `simulate_intervention(variable, value)` implements the do-operator with Monte Carlo simulation and returns effect estimates with confidence intervals; `rank_drivers()` orders all variables by causal effect on the target; `optimize_policy()` searches intervention settings under user constraints; counterfactuals can be generated conditional on observed evidence.
- **Validation.** `validate()` produces confounding bounds at varying assumed strengths, placebo tests, and multiple refutation methods, summarizing how fragile each conclusion is to violations of the causal assumptions.
- **Experimentation.** `ExperimentAnalysis` estimates treatment effects from randomized or quasi-experimental data, including a doubly-robust estimator, and analyzes effect heterogeneity across covariates.
- **Applied modules.** Uplift modeling with segmentation into persuadable, sure-thing, lost-cause, and sleeping-dog cohorts; marketing attribution via causal Shapley values with constrained budget optimization.

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References

Bang, H., & Robins, J. M. (2005). Doubly robust estimation in missing data and causal inference models. *Biometrics*, *61*(4), 962–973. https://doi.org/10.1111/j.1541-0420.2005.00377.x

Battocchi, K., Dillon, E., Hei, M., Lewis, G., Oka, P., Oprescu, M., & Syrgkanis, V. (2019). *EconML: A python package for ML-based heterogeneous treatment effects estimation.*

https://github.com/py-why/EconML

Chen, H., Harinen, T., Lee, J.-Y., Yung, M., & Zhao, Z. (2020). CausalML: Python package for causal machine learning. *arXiv Preprint arXiv:2002.11631*. https://doi.org/10.48550/arXiv.2002.11631

Cranmer, K., Brehmer, J., & Louppe, G. (2020). The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, *117*(48), 30055–30062. https://doi.org/10.1073/pnas.1912789117

Gutierrez, P., & Gérardy, J.-Y. (2017). Causal inference and uplift modelling: A review of the literature. *Proceedings of The 3rd International Conference on Predictive Applications and APIs, PMLR*, *67*, 1–13.

Pearl, J. (2009). *Causality: Models, reasoning and inference* (2nd ed.). Cambridge University Press. https://doi.org/10.1017/CBO9780511803161

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Sharma, A., & Kiciman, E. (2020). DoWhy: An end-to-end library for causal inference. *arXiv Preprint arXiv:2011.04216*. https://doi.org/10.48550/arXiv.2011.04216

Shimizu, S., Hoyer, P. O., Hyvärinen, A., & Kerminen, A. (2006). A linear non-Gaussian acyclic model for causal discovery. *Journal of Machine Learning Research*, *7*, 2003–2030.

Zheng, X., Aragam, B., Ravikumar, P., & Xing, E. P. (2018). DAGs with NO TEARS: Continuous optimization for structure learning. *Advances in Neural Information Processing Systems*, *31*. https://doi.org/10.48550/arXiv.1803.01422